

LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
Observation Task No.	1
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Date	30-08-2026

1. TITLE

Design and Implementation of a College Information Portal Using HTML5

2. AIM

To develop a structured College Information Portal using HTML5 elements including semantic sections, text formatting, hyperlinks, ordered and unordered lists, tables, images, registration form controls, and audio/video multimedia.

3. OBJECTIVE

1. To understand the structure of a complete HTML5 webpage.
2. To use semantic elements such as header, nav, section, and footer.
3. To implement formatted text and mathematical/chemical expressions using sup and sub.
4. To display student information using HTML tables.
5. To create a registration form using different HTML5 input types.
6. To embed images and multimedia audio/video elements.
7. To implement hyperlinks to internal and external web resources.

4. THEORY

HTML5 is a markup language used to structure content on web pages. Semantic elements such as **header**, **nav**, **section**, and **footer** provide meaningful organization. HTML provides formatting tags such as b, i, u, sup, and sub, as well as lists and tables for structured information. Forms support controls including text, email, password, radio buttons, checkboxes, select menus, textarea, date, color, and range. The audio and video elements provide native multimedia playback controls.

Application Flow: Browser → HTML5 Document → Header & Navigation → Information Sections → Table & Form → Image & Multimedia → Footer

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and set the title as 'University Information Center'.
2. Create a header containing the institute name and introductory message.
3. Create navigation links for Home, Campus, Official Website, and Search.
4. Create an information section containing formatted text and formulas.
5. Add ordered and unordered lists for academic programs and campus facilities.
6. Construct a student information table with Student ID, Name, Department, and Email.
7. Add a campus image using the img element.
8. Create an admission form with text, email, password, radio, checkbox, select, and textarea controls.
9. Add date, color, and range input controls to the form.

10. Add submit and reset buttons.
11. Embed audio and video elements using the controls attribute.
12. Add a footer with copyright and contact information.
13. Open the webpage in a browser and verify all elements.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>University Information Center</title>
</head>

<body>

<header>
  <h1>XYZ Institute of Technology</h1>
  <p>Welcome to the University Information Center.</p>
</header>

<nav>
  <a href="#">Home</a> |
  <a href="#">Campus</a> |
  <a href="https://www.google.com" target="_blank">Official Website</a> |
  <a href="https://www.google.com" target="_blank">Search</a>
</nav>

<hr>

<section>
  <h2>About Our Institute</h2>

  <p>
    <b>XYZ Institute</b> provides <i>Excellent Education</i>
    with <u>Advanced Facilities</u>.
  </p>

  <p>
    A<sup>2</sup> + B<sup>2</sup> = C<sup>2</sup><br>
    C<sub>2</sub> is Carbon Dioxide.
  </p>

  <h3>Academic Programs</h3>
  <ol>
    <li>Computer Science</li>
    <li>Electronics</li>
    <li>Mechanical</li>
  </ol>

  <h3>Campus Facilities</h3>
  <ul>
    <li>Computer Labs</li>
    <li>Library</li>
    <li>Playground</li>
  </ul>
</section>

<hr>

<section>
  <h2>Student Information</h2>

  <table border="1">
    <tr>
      <th>Student ID</th>
      <th>Name</th>
      <th>Department</th>
      <th>Email</th>
    </tr>

    <tr>
      <td>201</td>
      <td>Arjun</td>
      <td>CSE</td>
      <td>arjun@gmail.com</td>
    </tr>

    <tr>
      <td>202</td>
      <td>Priya</td>
      <td>ECE</td>
      <td>priya@gmail.com</td>
    </tr>
  </table>
</section>

<hr>

<section>
  <h2>Campus Image</h2>
  
</section>

<hr>

<section>
  <h2>Admission Form</h2>
```

```

<form>
  Name:
  <input type="text">
  <br><br>

  Email:
  <input type="email">
  <br><br>

  Password:
  <input type="password">
  <br><br>

  Gender:
  <input type="radio" name="gender"> Male
  <input type="radio" name="gender"> Female
  <br><br>

  Interests:
  <input type="checkbox"> Coding
  <input type="checkbox"> Music
  <br><br>

  Department:
  <select>
    <option>CSE</option>
    <option>ECE</option>
    <option>EEE</option>
    <option>Mechanical</option>
  </select>
  <br><br>

  Address / Comments:
  <br>
  <textarea rows="3" cols="25"></textarea>
  <br><br>

  Admission Date:
  <input type="date">
  <br><br>

  Favourite Color:
  <input type="color">
  <br><br>

  Skill Level:
  <input type="range">
  <br><br>

  <input type="submit">
  <input type="reset">
</form>
</section>

<hr>

<section>
  <h2>Media Center</h2>

  <audio controls>
    <source src="audio.mp3" type="audio/mpeg">
  </audio>

  <br><br>

  <video width="350" controls>
    <source src="campus.mp4" type="video/mp4">
  </video>
</section>

<footer>
  <p>© 2026 XYZ Institute | contact@xyz.edu.in</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Code / Syntax	Explanation
<header>, <nav>, <section>, <footer>	Semantic HTML5 elements used to organize webpage content.
	Creates hyperlinks to other pages or external websites.
, <i>, <u>	Applies bold, italic, and underline formatting.
<sup>, <sub>	Displays superscript and subscript text for formulas.
, ,	Creates numbered and bulleted lists.
<table>, <tr>, <th>, <td>	Creates structured student information in rows and columns.
<input>	Provides different form controls such as text, email, password, date, color, and range.
<select>, <option>	Creates a drop-down menu for department selection.
<textarea>	Provides a multi-line area for comments or address.
<audio controls>, <video controls>	Embeds multimedia with browser playback controls.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open index.html	Header, navigation, and sections rendered	Rendered correctly	Pass
TC-02	Verify formatting and lists	Formatting, formulas, and lists display properly	Display correctly	Pass
TC-03	Inspect student table	Table displays four columns and two rows	Table rendered correctly	Pass
TC-04	Fill admission form	Inputs accept text, selection, and controls	Form controls work	Pass
TC-05	Click audio/video controls	Media players display browser controls	Controls displayed	Pass

9. OUTPUT

Output 1: University Information Center displaying header, navigation, formatted information, academic programs, facilities, and student table.

Output 2: Campus Image, Admission Form, Audio and Video controls displayed successfully.

XYZ Institute of Technology Welcome to the University Information Center.	
Home Campus Official Website Search	
About Our Institute Excellent Education with Advanced Facilities.	
Academic Programs 1. Computer Science 2. Electronics 3. Mechanical	
Student Information 201 - Arjun - CSE - arjun@gmail.com 202 - Priya - ECE - priya@gmail.com	

Admission Form

Name: _____ Email: _____

Gender: ☐ Male ☐ Female

Department: [CSE ▼]

[Submit] [Reset]

Media Center

Audio Player

Video Player

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10. RESULT

Thus, the University Information Center was successfully implemented using HTML5. The webpage was verified for semantic structure, hyperlinks, text formatting, lists, student table, image, registration form controls, and multimedia elements.